SMART INDIA HACKATHON 2026

JalRakshak

Your Water. Your Village. Your Voice.

- **Problem Statement ID -** SIH26255
- **Problem Statement Title -** Development of a solution using AI and ML to monitor the functionality of FHTC in real time at both the Gram Panchayat and household levels. This solution should provide alerts on water quality issues, allow village members to give feedback through mobile devices, and automatically sync data with IMIS/Sujal Gaon IDs.
- **Organization -** Ministry of Jal Shakti (MoJS)
- **Department -** National Jal Jeevan Mission (NJJM), Department of Drinking Water & Sanitation (DDWS)
- **Theme -** Clean & Green Technology
- **PS Category -** Hardware
- **Team ID -** [enter from portal]
- **Team Name -** Team Vistaar



Team Vistaar

THE PROBLEM & OUR SOLUTION

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THE PROBLEM

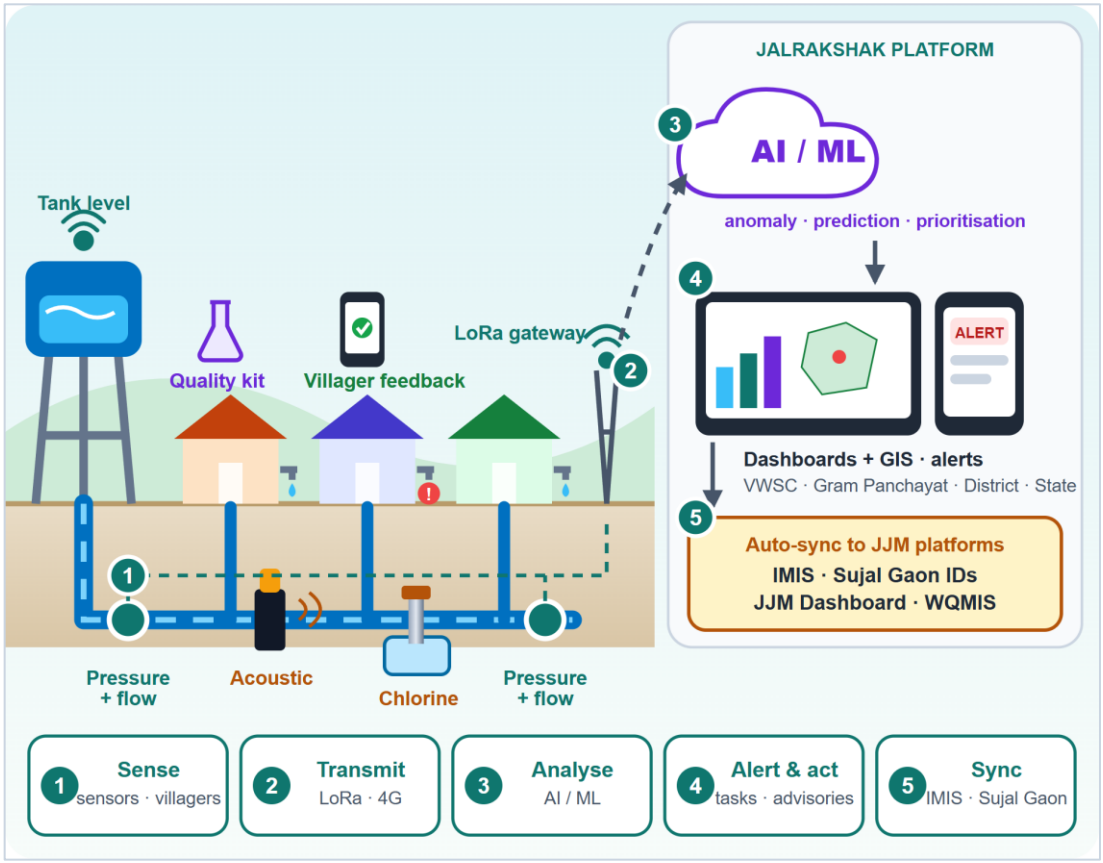
Periodic reporting only
Monitoring relies on periodic reports and field inspections, so disruptions are found late

Disruptions go unseen
Non-functional taps, intermittent supply, poor water quality, inadequate pressure, infrastructure failures

Limited community feedback
No real-time way for villagers to report, or to validate service

No unified platform
Sensors, mobile apps, water-quality systems and citizen feedback are not brought together

Taps installed, service uncertain
98.1% of surveyed households had a tap, but only 76.0% were assessed as functional [1]



OUR SOLUTION: JALRAKSHAK

1. **Continuous FHTC monitoring**
   at Gram Panchayat and household levels using AI/ML
2. **Real-time alerts**
   water supply disruption, water quality deterioration, low pressure, leakages, infrastructure failure
3. **Community feedback**
   complaints, photographs and geo-tagged observations from mobile devices, in local languages
4. **AI/ML analytics**
   abnormal service patterns, failure prediction, maintenance prioritisation
5. **Automatic sync**
   monitoring data, alerts and feedback to IMIS, Sujal Gaon IDs and other JJM platforms
6. **Interactive dashboards**
   GIS visualisation, service status indicators, analytics for multiple administrative levels
7. **Scalable, interoperable, secure**
   data security and ease of deployment

SENSE Nodes + villagers → **UNDERSTAND** Rules + AI/ML → **ALERT** Alert Center → **REPORT** IMIS / JJM platforms → **FIX** Task + repair → **VERIFY** Readings recover

DEMO VIDEO (YouTube)
[add link after upload]

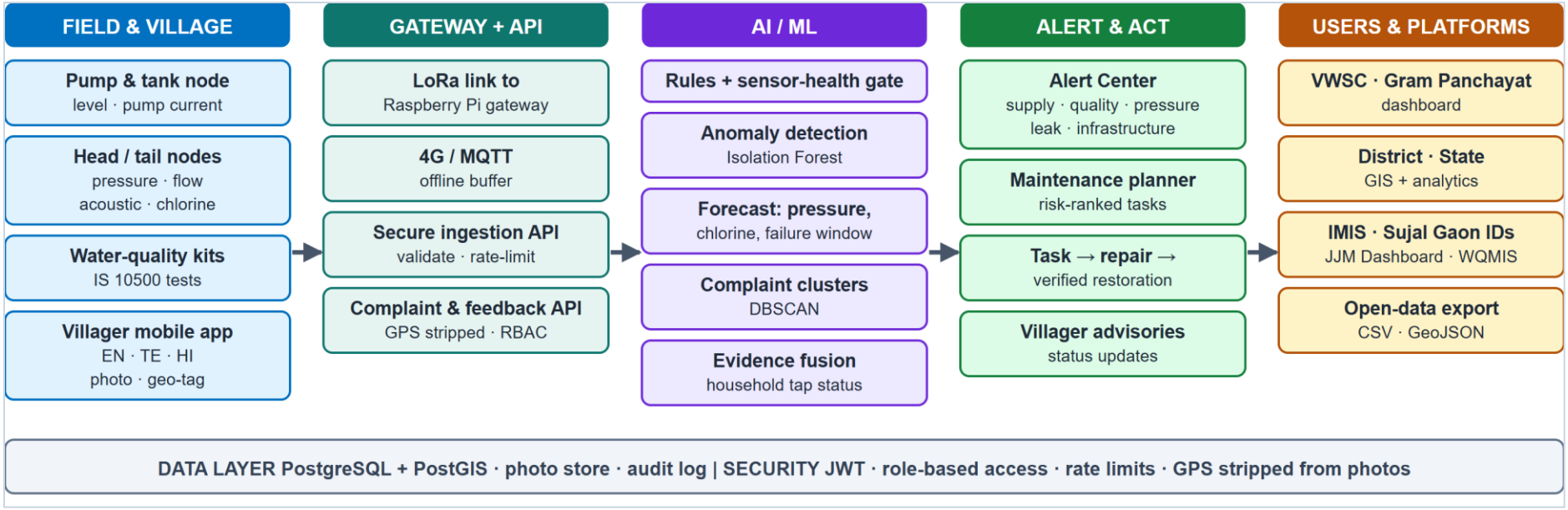
For: Village Water & Sanitation Committees (VWSCs), Gram Panchayats, implementing agencies and State authorities.

[1] DDWS, Functionality Assessment of Household Tap Connection, National Report 2024 (weighted sample estimates).

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TECHNICAL APPROACH

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Technology stack

App: React, Vite, Tailwind CSS, MapLibre, i18next (EN / TE / HI)

Backend: Python, FastAPI, SQLAlchemy, JWT

Database: PostgreSQL + PostGIS (SQLite for development)

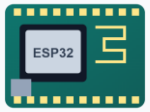
AI/ML: scikit-learn, NumPy: Isolation Forest, DBSCAN, Theil-Sen trend

IoT: ESP32 (MicroPython firmware), ADS1115, LoRa SX1276, Raspberry Pi gateway, MQTT

Interfaces: REST + OpenAPI, GeoJSON, CSV

Deployment: Docker, Nginx

Hardware we use and how it connects (circuit diagram on the next slide)



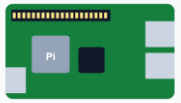
ESP32 node

Reads sensors through ADC, I2C and GPIO; deep sleep between samples



LoRa SX1276

SPI radio, 865-867 MHz, node to village gateway



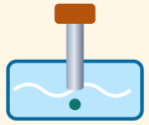
Pi gateway

Buffers readings, sends over 4G MQTT to the ingestion API



Acoustic sensor

Piezo clamp on the pipe; leak noise is amplified, filtered and analysed on the node



Chlorine sensor

4-20 mA probe in a flow cell at the tail end; measured mg/L, low-chlorine alert



Pressure x2

4-20 mA transmitters, head and tail; low-pressure alert



Flow meter x2

Pulse output, head and tail; head-to-tail gap means a leak



Ultrasonic level

JSN-SR04T on the tank; supply and tank health



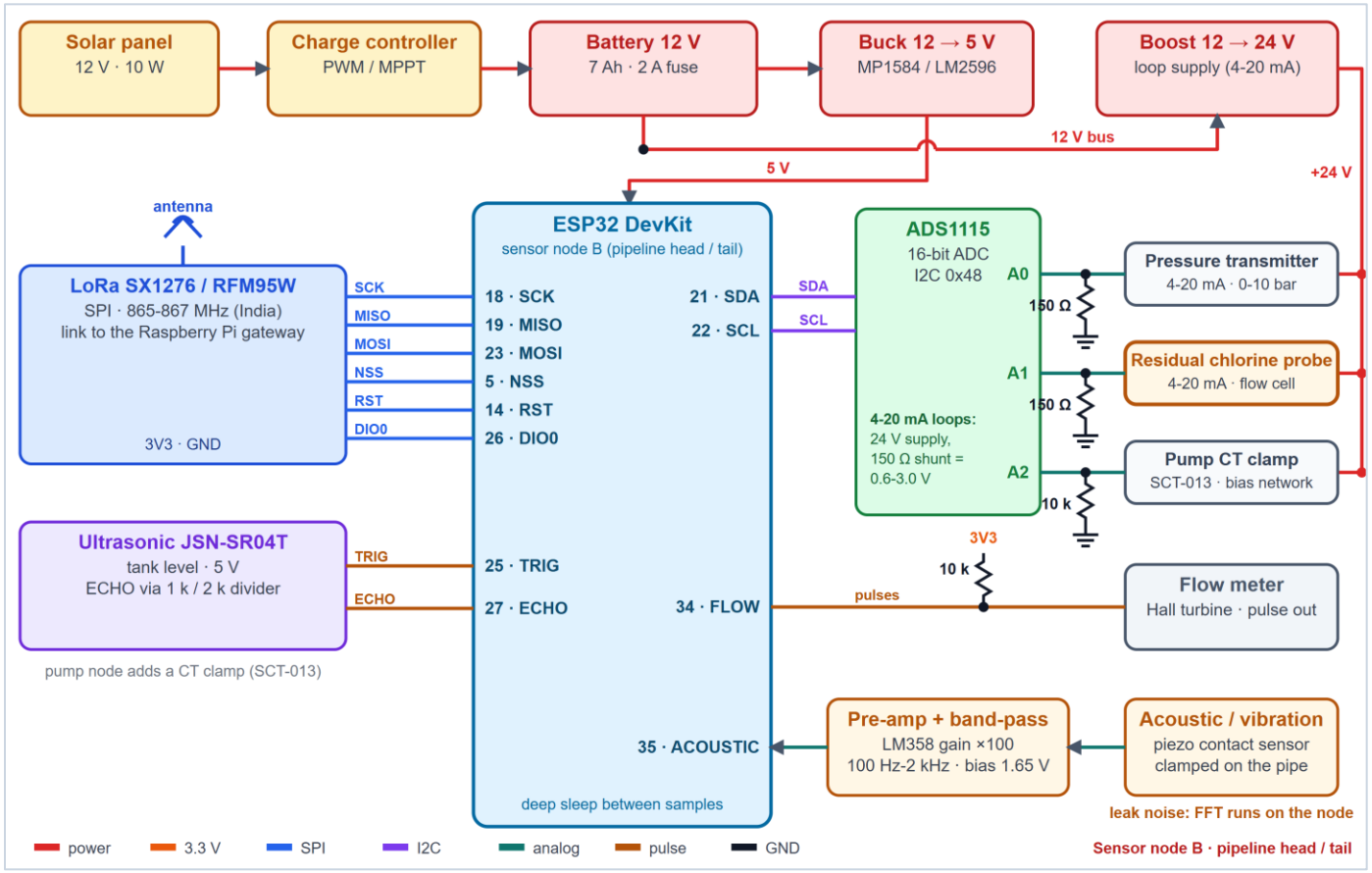
Solar + battery

12 V 10 W panel, 7 Ah battery, IP65 box; buck and boost rails

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FEASIBILITY AND VIABILITY

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Circuit of sensor node B (pipeline head / tail): ESP32 with LoRa, ADS1115, 4-20 mA loops, flow, acoustic and tank-level inputs.

Scalability

- One village, then a cluster per gateway, block, district, state
- Stateless APIs; PostgreSQL + PostGIS
- Same node design in every village

Interoperability

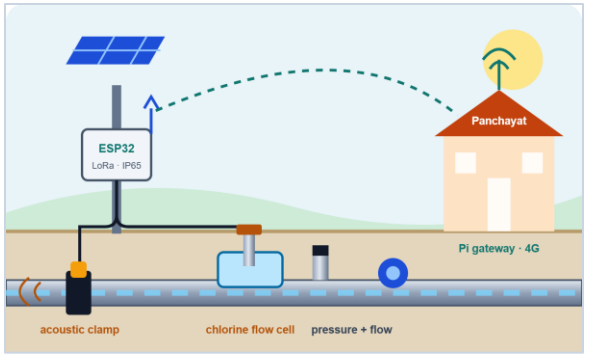
- Keyed to Sujal Gaon and Sujalam Bharat IDs
- Sync with IMIS, JJM Dashboard, WQMIS
- Open formats: REST/OpenAPI, GeoJSON, CSV

Data security

- Role-based access by village; hashed passwords
- Rate limits, audit log, GPS removed from photos
- No personal data sent to other platforms

Ease of deployment

- Solar node, plug-in LoRa gateway, no grid cabling
- Mobile web app: nothing to install
- Docker deployment, one configuration file



Node installation concept

- Solar-powered pole node with an IP65 box
- Clamp-on acoustic sensor: no digging
- Chlorine flow cell at the tail end
- LoRa to the Panchayat gateway

Feasibility analysis

- Off-the-shelf sensors, ESP32, LoRa, Raspberry Pi
- Node firmware and gateway written; 278 automated tests pass
- Open-source software; rules work from day one, AI/ML improves with data

Potential challenges and strategies to overcome them

False alarms: persistence windows; HIGH needs two kinds of evidence

Sensor fault or drift: health gate; chlorine probe checked with a DPD kit

Rural network: gateway buffers readings, sends when online

Trust in AI: every score shows its reasons

Leak vs noise: per-pipe acoustic baseline plus flow gap

Platform access: adapters ready; official access and specifications needed

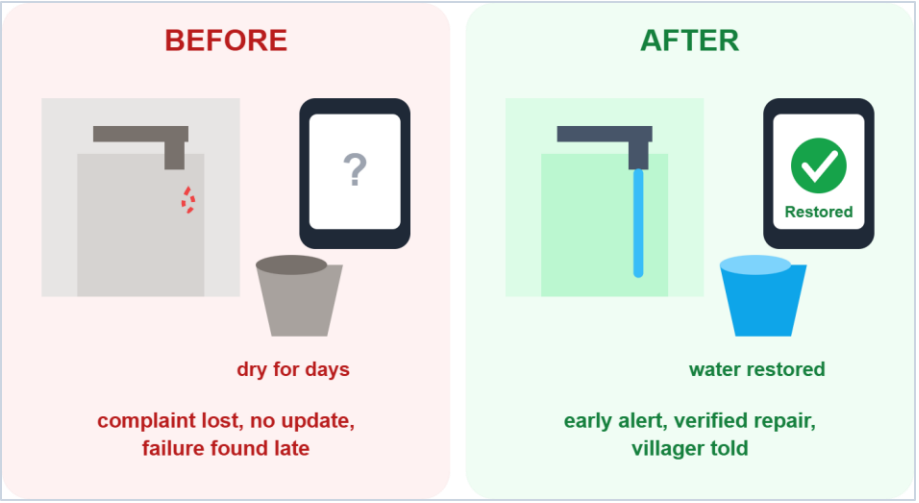
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IMPACT AND BENEFITS

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Expected outcomes

1. **AI/ML-enabled platform:** real-time monitoring of Functional Household Tap Connections and rural water supply service delivery
2. **Early identification and prediction:** of service disruptions, water quality issues and infrastructure failures, enabling timely corrective action
3. **Community participation:** mobile-based citizen feedback, grievance reporting and service validation
4. **Automatic integration:** of monitoring data with IMIS, Sujal Gaon IDs and other Jal Jeevan Mission digital platforms
5. **Operational efficiency:** preventive maintenance planning, transparency and evidence-based decision-making
6. **Reliability, accountability, sustainability:** of FHTCs through continuous AI-assisted monitoring and analytics



Who benefits: Village Water & Sanitation Committees, Gram Panchayats, implementing agencies, State authorities and villagers

VWSC / Gram Panchayat

Before: Learns of failures late; no priority order. Only 55.2% of villages had a VWSC [1]

After: Alert Center and a ranked inspection list; proof that a repair worked

Implementing agencies

Before: Periodic inspections and reactive repairs

After: Preventive maintenance plan with reasons; repeat-failure hotspots

State authorities

Before: Coverage numbers, not service quality

After: Multi-level analytics and GIS maps; exports for evidence-based decisions

Villagers

Before: Complaint disappears; the tap may be dry for days

After: Report in own language; told when water is truly restored; quality advisories

Roadmap

Idea submission → Working software platform → Bench rig with real nodes → Finale demo → Field pilot

Pilot will measure: time to detect, time to repair, share of taps working, complaints closed with verified restoration.

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RESEARCH AND REFERENCES

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Details / links of the reference and research work

[1] DDWS, Functionality Assessment of Household Tap Connection, National Report 2024
jaljeevanmission.gov.in/sites/default/files/2025-12/FHTC_Na...

[2] PIB, 13 Aug 2026: 75% of rural habitations covered with Sujal Gaon IDs
pib.gov.in/PressReleasePage.aspx?PRID=2298959

[3] PIB, 30 Dec 2025: Jal Seva Aankalan launched
pib.gov.in/PressReleasePage.aspx?PRID=2209798

[4] Cabinet, Mar 2026: JJM extended to Dec 2028, service-delivery focus
pmindia.gov.in/en/news_updates/

[5] JJM Dashboard (coverage, GIS, IoT pilot view)
ejalshakti.gov.in/JJMReport/JJMIndia.aspx

[6] JJM-WQMIS (laboratory and FTK water-quality testing)
ejalshakti.gov.in/WQMIS/

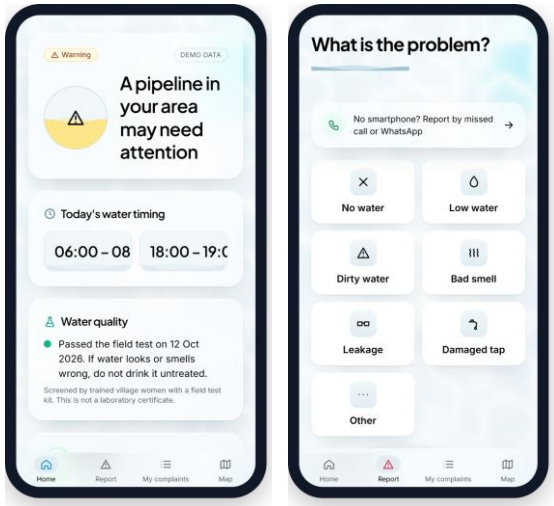
[7] PIB, 31 Mar 2021: JJM deploys sensor-based IoT devices (off-grid pilot)
pib.gov.in/PressReleaseIframePage.aspx?PRID=1708701

[8] Jal Jeevan Samvad, Oct 2023: IoT water-quality monitoring with complaint tracking
jaljeevanmission.gov.in/sites/default/files/2025-05/JalJeev...

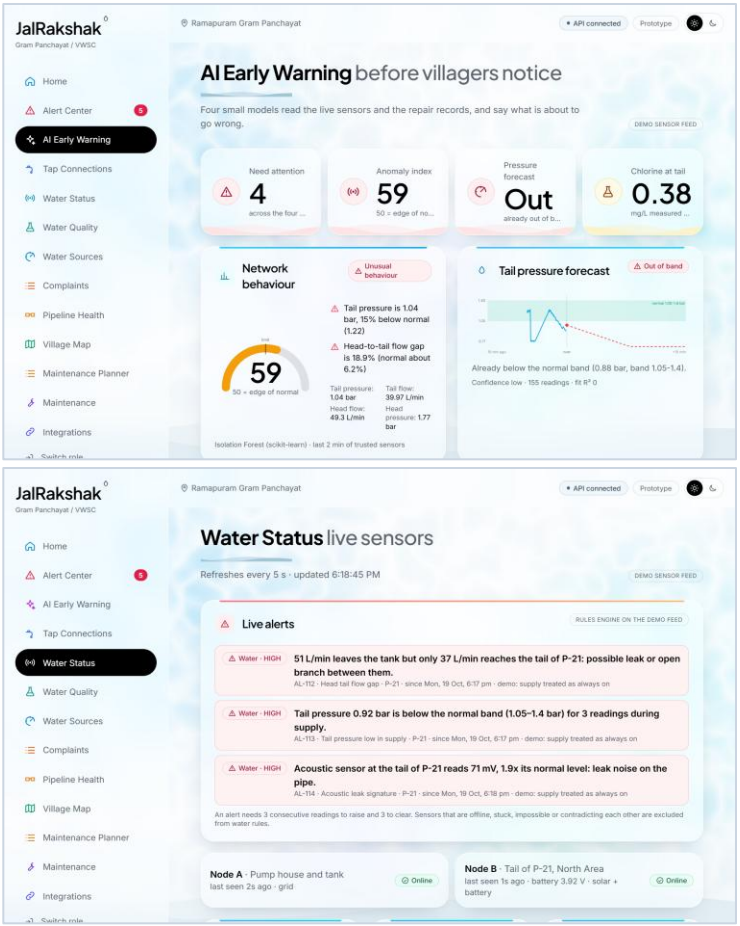
[9] Systematic review: ML for anomaly detection in smart water metering networks (2024)
sciencedirect.com/science/article/pii/S1474706524000160

[10] SIH 2026 problem statement SIH26255, Ministry of Jal Shakti
sih.gov.in/sih2026PS

Working platform screens



Villager app: tap status and water quality (left); report a problem with photo and opt-in location (right)



Gram Panchayat: AI Early Warning with measured chlorine (top right) and Water Status with acoustic leak alerts (bottom right).

Demo video: [add YouTube link]

Source code: private repository, access on request (includes node firmware, gateway and an ESP32 simulation project)

pH, TDS and turbidity readings are screening indicators that trigger confirmatory laboratory or field-kit testing; they are not a safety certification.